MOHAMED H. MOUSA .PhD

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EDUCATION

Ph.D., Engineering (Computational Neuroscience) 2022

Reverse Engineering Spinal Motoneuron Properties in Slice and Whole Cord Preparations Using Computational Modeling

Advisors: Dr. Sherif Elbasiouny
Dept. Biomedical, Industrial, and Human Factors Engineering,
Wright State University, OH, USA

M.Sc. Biomedical Engineering 2017

Computer Modeling of Calcium & Potassium Channels Dendritic Distributions in Spinal Motoneurons

Advisors: Drs. Ahmed Hesham and Sherif Elbasiouny
Dept. of Systems & Biomedical Engineering,
Cairo University, Cairo, Egypt

B.Sc. Systems & Biomedical Engineering 2013

GPA: 3.94, Ranked 1'st (105 students)

Dept. of Systems & Biomedical Engineering,
Cairo University, Cairo, Egypt

ACADEMIC EXPERIENCE

Postdoctoral Fellow Research Associate 2024- present

Advisor: Christopher Thompson, PhD
Healthy and Rehabilitation Sciences, College of Public Health
Temple University, PA, USA

Activities:

- Collecting electrophysiological data from humans, and animal models
- Developing frameworks to process, decompose, analyze electrophysiological data using MATLAB, Python, R and Simulink.
- Analyzing the non-linear dynamics of spinal networks using machine learning and network graph theory
- Developing and optimizing 3D biophysical computational models of spinal motoneurons and interneurons, simulating neuron discharge patterns, connectivity changes post- spinal cord injury

Postdoctoral Fellow 2023 - 2024

Advisor: Sherif Elbasiouny, PhD
Department of Neuroscience, Cell Biology, and Physiology (NCBP),
Wright State University, OH, USA

Activities:

- Leading clinical study, to develop a diagnostic biomarker for ALS by enrolling, consenting conducting EMG recordings.

- Analyzing and reverse-engineering of electrophysiological data from in-vivo animals' recordings using computational models.
- Conducting statistical analyses, and scientific visualizations to direct experiments and provide insightful conclusions

PhD Candidate 2017 - 2022
Advisors: Dr. Sherif Elbasiouny
Department of Biomedical, Industrial, and Human Factors Engineering,
Wright State University, OH, USA
Activities:

- Developed high-fidelity computational models of motoneurons, to reverse engineer experimental electrophysiological data.
- Automated analyses, feature extraction of electrophysiological data, using Python and MATLAB.
- Developed mathematical models for ionic channels to improve the computational model's accuracy and enable the replication of complex motoneurons behaviors in-silico

AWARDS

- Excellence award as the top Ph.D. graduate student in biomedical engineering, WSU, USA 2023
- Science Gateways Community Institute (SGCI) Science Ambassador - $1200 Travel award 2019
- Travel Award by Professional Development Grant, WSU, USA 2018
- Best paper award, Cairo International Biomedical Engineering (CIBEC) 2016
- Graduated First in Class with Distinction, Cairo University, Egypt 2013
- Recipient of annual Excellence Award (top 5%), Cairo University, Egypt 2009-2012

SCHOLARSHIPS

- NIH BRAIN Initiative Summer Course — *Models and Neurobiology* , University of Missouri-Columbia, USA June 2017
  Interdisciplinary computational neuroscience training covering neural modeling and biological systems
- Research Assistant Scholar, Neuro Engineering, Rehabilitation & Degeneration Laboratory, Wright State University, OH, USA Summer 2015, 2016
  Simulated patch cellular properties in motoneurons using NEURON and NMODL
- Undergraduate Research Assistant, Hochschule Zittau/Görlitz, Germany Summer 2012
  Heat transfer modeling and simulations using MATLAB and fuzzy logic theory

TEACHING EXPERIENCE

Departments of Neuroscience, Cell Biology, & Physiology (NCBP) and
Biomedical, Industrial, & Human Factors Engineering (BIE)
Wright State University, Dayton OH, USA

Courses taught as teaching assistant:

From neurons to behavior – in health and Disease (BME 7380/BMS 9910) 2017 - 2024
Computational Neuroscience (NEU 4600/PN 6600) 2021 - 2024
Design and Analysis of Engineering Experiments (EGR 7050) 2018

Biomedical & Systems Engineering Dept.,
Cairo University, Giza, Egypt.

Courses taught as teaching assistant:

Classic Control Systems. (*BME sophomore , class size: 78 students)*	2016
Medical Equipment – MRI (*BME senior, class size: 62 students*)	2017
Medical Equipment – X-Ray (*BME junior, class size: 66 students*)	2015
Data Structure & Algorithms using C (*BME sophomore, class size: 76 students*)	2015
Computer Vision (*BME senior, class size: 66 students*)	2016
Biomedical Electronics (*BME junior, class size: 62 students*)	2017

Volunteer to teach summer school.

Programming under UNIX shell in C	2016
Medical equipment (radiology)	2015

TECHNICAL SKILLS:

- Machine Learning
- Modeling & Simulations
- Signal & Image processing
- Network & Graph Theory
- Statistical Analysis
- Clinical Management
- Nonlinear Dynamics

PROGRAMMING EXPERIENCE

- Python
- Matlab
- NEURON
- C/C++
- Hoc, NMODL
- JavaScript
- Scipy
- SAS
- Pingoiun
- statsmodels
- Pandas
- SQL
- NumPy
- Matplotlib
- Seaborn
- Plotly
- PyTorch

PROFESSIONAL MEMBERSHIPS

Member, The American Physiological Society	2020 - present
Member, Society for Neuroscience (SfN)	2015 - present
Member, The Syndicate of Engineers (Egypt), Electrical Engineering	2013 – 2016
Member, IEEE Engineering in Medicine and Biology Society (EMBS)	2017
Member, International Association for the Exchange of Students for Technical Experience (IAESTE)	2012

GRANTS & FELLOWSHIPS

Milton Safenowitz Postdoctoral Fellowship (2023 ,2024)

- Budget : $150,000
- Agency: The ALS Association, USA
- Project: Identifying early biomarkers of ALS via human sEMG and computational modeling
- Status: Letter of Intent (LOI) Approved; Full Proposal: Not Funded

LIST OF PUBLICATIONS AND PODIUM TALKS

I- PEER-REVIEWED PUBLICATIONS

1) Abdul Halim, I., Gerber, K. R., Gelford, W. B., Ellington, C. L., **Mousa, M. H**., Garrett, T. L., & Elbasiouny, S. M. (2026). SK channel upregulation and sex-specific mechanisms drive spinal motoneuron reduced excitability with age. *Frontiers in Aging Neuroscience*

2) Zaback, M., **Mousa, M. H.**, & Mildren, R. L. (2025). Shining light on spinal co-ordination: internal states shape interneuron recruitment of muscle synergies. *The Journal of Physiology*.

3) **Mousa, M. H**. & Elbasiouny, S. M. (2025). A Multi-Scale, High-Fidelity Computational Model of the Mouse Triceps Surae Motor Pool. *47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*

4) **Mousa, M. H.,** Wages, N. P., & Elbasiouny, S. M. (2024). Onion skin is not a universal firing pattern for spinal motoneurons: simulation study. *Journal of neurophysiology*, *132*(1), 240-258.

5) Wages, N. P., **Mousa, M. H**., Clark, L. A., Tavoian, D., Arnold, W. D., Elbasiouny, S. M., & Clark, B. C. (2024). Reductions in motor unit firing are associated with clinically meaningful leg extensor weakness in older adults. *Calcified Tissue International*, *114*(1), 9-23.

6) Moustafa, M., **Mousa, M. H**., Saad, M. S., Basha, T., & Elbasiouny, S. M. (2023). Bifurcation analysis of motoneuronal excitability mechanisms under normal and ALS conditions. Frontiers in Cellular Neuroscience, 17, 1093199.

7) Gamal M, **Mousa MH**, Eldawlatly S, Elbasiouny SM (2021) In-silico development and assessment of a Kalman filter motor decoder for prosthetic hand control. *Comput Biol Med.* 132, 104353

8) **Mousa MH** and Elbasiouny SM (2021) Estimating the effects of slicing on the electrophysiological properties of spinal motoneurons under normal and disease conditions. *J Neurophysiology* 124(4): 1450-67

9) **Mousa MH** and Elbasiouny SM (2020) Dendritic distributions of L-type Ca^{2+} and SK_L channels in spinal motoneurons: A simulation study. *J Neurophysiology*, 124(4):1285-1307

10) Abdelaal AY, **Mousa M**, Gamal M, Elbasiouny SM, Eldawlatly S (2020). A Classification Approach to Recognize the Firing of Spinal Motoneurons in Amyotrophic Lateral Sclerosis. 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), Montreal, QC, Canada, 2020, pp. 3680-3683.

11) Elbasiouny SM and **Mousa M** (2020) Morphologically-detailed Cellular and Pool Motoneuron Models. *In: Encyclopedia of Computational Neuroscience (Jaeger D, Jung R, eds)*. Springer-Verlag Berlin Heidelberg: Springer.

12) Mahrous AA, **Mousa MH**, Elbasiouny SM (2019) The Mechanistic Basis for Successful Spinal Cord Stimulation to Generate Steady Motor Outputs. *Frontiers in Cellular Neuroscience* 13:359-374

13) Gamal M, **Mousa M**, Eldawlatly S, Elbasiouny SM (2018). Automated Cell-Type Classification and Death-Detection of Spinal Motoneurons. 2018 9th Cairo International Biomedical Engineering Conference (CIBEC), Cairo, Egypt, IEEE, pp. 57-60.

14) **Mousa MH**, Kandil AH, and Elbasiouny SM (2016). "Simulation of dendritic L-type Ca channels' warm-up phenomenon in spinal motoneurons", *8th Cairo International Biomedical Engineering Conference (CIBEC)*, Cairo, Egypt, December 2016, pp. 31-34: IEEE.

II- PODIUM TALKS

"Activity and Functional Connectivity of Lumbar Spinal Interneurons at Rest: Sparse Yet Structured" Neural Sciences Research Day, Temple University Lewis Katz School of Medicine. Philadelphia December 2025

"Introduction to Computational Modeling", Ministry of Communications and Information Technology, NAID's assistive technologies internship program 2025 – Egypt - **Webinar Invited Speaker**

"Resting State Activity of Lumbar Spinal Interneurons: Keeping the Motoneurons Warm through Distributed and Functionally Coupled Subpopulations" The 14th International Motoneuron Society Meeting, July 8-11, St. John's, Newfoundland, Canada

"The role of computational models in experimental research: predictive vs. explanatory power" Neuroscience Gateway Satellite Workshop, George Washington University, D.C November 2023 – **Invited Talk**

"Understanding the Dichotomy in Motoneuron Firing Patterns: "Onion Skin" vs "AHP"" Wright state University Research Celebration, Dayton, OH October 2023

"Differences between SK2 and SK3 currents in mouse spinal motoneurons: A simulation study", Society for Neuroscience. Chicago, IL November 2019

"Dendritic distribution of Cav1.3 and SK channels in spinal motoneurons: a simulation study". The International Motoneuron Meeting, Boulder, CO, June 2018

"A Review of Below-Knee Socket Materials Using Finite Element Analysis" The 14th Annual Dayton Engineering Sciences Symposium October 2018

SERVICE & LEADERSHIP

- Program Chair, IEEE EMBS International Student Conference – ISCegypt (2017)
- Chair, IEEE BME Cairo University Chapter (2017)
- Instructor, "C Programming Under Linux Shell" summer workshop (2016)
- Coordinator, BME Project Development Competition, Cairo University IEEE EMBS Chapter (2015)